Deterministic Caging of Probabilistic Large Language Models: Architecture, Field Receipt, and a Drift-Audit Case Study

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Thesis community: This work is part of the Deterministic Caged AI (DCAI) research line: https://zenodo.org/communities/dcai

Companion receipts: First Documented Caging of an LLM — DOI 10.5281/zenodo.23265723. Substrate Determinism Receipt 1 (SDR-1) — ledger of record at github.com/LLong2026/four-horsemen.

--- **AMENDMENT EDITION v1.1 (October 10, 2026):** This edition appends the tri-vendor field extension (SDR-CL1, SDR-CL2) to the v1.0 paper below. The v1.0 text is preserved unmodified; the new material begins at the section titled "AMENDMENT (v1.1)" at the end of this document. Version history on this DOI preserves v1.0. Two-key curation: drafted and vetted by the operator agent (Gabriel), published on the author's explicit GO (October 10, 2026).

Abstract

Large language models are probabilistic systems whose outputs cannot, in general, be guaranteed correct. The industry's dominant responses — alignment tuning, evaluation benchmarks, constitutional prompting — attempt to make the model itself more reliable, and remain probabilistic. This paper documents a different approach: **deterministic caging**, in which a probabilistic language model is placed inside a deterministic mathematical boundary that checks every state-changing output against preregistered, verifiable ground truth. The cage does not attempt to make the model right; it makes the model's errors *detectable, bounded, and receipted*.

We present three contributions. First, the architecture: a bifurcated governance stack in which a probabilistic layer proposes and a deterministic layer validates, with Substrate Determinism Receipts (SDRs) providing byte-identical, replayable audit records of every governed action. Second, a field receipt: the first documented caging of a live LLM (October 8–9, 2026), in which a model-pinned agent produced a drift detection with a cryptographic receipt that withstood direct third-party scrutiny, and which was published with an open trail. Third, and most unusually, this paper audits its own genesis: an AI-generated draft of this very paper exhibited textbook confabulation — fabricated evaluation data, fabricated model attribution, and a fabricated institutional identity — and the audit of that draft is included here as a worked case study. The paper you are reading is thus both a description of the cage and an artifact produced inside it.

Keywords: deterministic caging, LLM governance, non-determinism, audit receipts, falsifiability, AI confabulation, bifurcated governance, post-quantum compliance

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1. Introduction

As financial value moves from visible checkouts into autonomous background software, the systems making consequential decisions increasingly embed probabilistic language models. The uncomfortable arithmetic is well known to information theory: a system that is right 99.9% of the time and executes a million times a day is wrong a thousand times a day. In settlement, minting, and infrastructure operations, the relevant question is not *whether* the model will err, but **what happens when it does**.

The alignment literature treats this as a model problem: make the model better. We treat it as a systems problem: **make the boundary deterministic**. A probabilistic component inside a deterministic boundary is not a liability; it is a caged computation whose every escape attempt is logged, whose every error becomes training data, and whose every permitted action carries a receipt that replays identically on demand.

This is the Cage Thesis of the SquirlOS research line: *probabilistic LLMs should be governed by a deterministic boundary cage that validates boundary outputs against ground truth, turning hallucinations into test data.*

This paper is deliberately small where its subject is grand. It makes exactly three claims, each with a public, inspectable receipt:

- A deterministic substrate can validate and receipt the actions of a probabilistic agent (architecture, §3–4).
- This has been done in the field, once, with the full trail published (field receipt, §5).
- The dominant failure mode of AI-assisted research — confident fabrication — is not hypothetical; it occurred in the drafting of this paper itself, and was caught by the same protocol the paper describes (case study, §6).

We claim nothing else. No field-wide validation, no production readiness, no guarantee of correctness for any deployment. One system, one event, full timestamped trail, publicly inspectable.

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2. Lineage and Prior Work

Science is a relay. This work stands on a lineage it did not invent and owes by name:

- **Nyquist (1924)** — established the mathematical limits of signal transmission regardless of channel design.
- **Hartley (1928)** — quantified information as a measurable quantity.
- **Shannon (1948)** — founded information theory; the probabilistic limits we cage within are his.
- **Kolmogorov (1965)** — defined descriptive complexity; the boundary between what must be stored and what can be regenerated is his.

The modern layer: Popper's falsifiability criterion (1959) supplies the epistemology — a governance claim is scientific only if a preregistered test can fail it. Determinism-by-architecture, rather than determinism-by-fiat, is the operating principle of the SquirlOS substrate.

Within the authors' own line, this paper is the synthesis of four prior artifacts:

- **Deterministically Governed Probabilistic Neural Computation** (provisional patent 64/119,191, filed July 25, 2026) — the method patent covering a probabilistic neural mesh governed as a mathematical

manifold under deterministic validation. The architecture in §3 operates under this filing.

- **The Caging Theorem** (Algebraic Collapse in Nondeterministic Language Models via Deterministic Substrate Governance, DOI 10.5281/zenodo.23067855) — the theoretical companion: a deterministic layer confines all finite algebraic combinations of LLM outputs inside a designed lower-dimensional structure; the cage is a theorem, not a metaphor.
- **SDR-1: Substrate Determinism Receipt 1** (September 24, 2026) — the freeze-then-replay receipt proving substrate determinism: identical canonical inputs produced identical checksum (0xc5f88dcf) across four independent instances, across replays, and across wall-clock windows; falsifier spec committed (d9cc042) *before* execution and the result appended (ef54c7b) after, in an append-only public ledger.
- **N10-ACK: First Receipted Human-in-the-Loop Critical Resolution** (DOI 10.5281/zenodo.23151356; sovereign briefing DOI 10.5281/zenodo.23152634) — the governance loop receipt: a critical anomaly detected, held, escalated to the human key, acknowledged, healed, and resolved with the full trail attached.

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3. The Caging Architecture

3.1 Bifurcated governance

The system is split into two layers with disjoint obligations:

- **The probabilistic layer** proposes. It contains the LLM agents: anomaly triage, healing recommendations, classification, drafting. It is permitted to be wrong. It is never permitted to *act* unvalidated.
- **The deterministic layer** validates and executes. It contains the constitution (a 12-article rule set expressed as enforceable invariants), the playbook engine, the circuit breakers, the rate limits, the tenant guards, and the receipt signer. Its outputs are functions of its inputs and its rules — replayable byte-identically.

The interface between the layers is the **boundary condition set**: every state-changing proposal from the probabilistic layer must pass explicit, preregistered checks before execution. A proposal that passes is executed and receipted. A proposal that fails is quarantined, logged, and — critically — *retained as drift data*.

3.2 Substrate Determinism Receipts (SDRs)

Every governed action produces an SDR: a canonical record of inputs, decision vector, outputs, and timestamps, hashed into the append-only ledger of record. Properties verified in SDR-1:

- **Parity** — identical inputs to independent instances yield identical outputs.
- **Replay** — the same instance healing the same batch twice yields identical decision vectors, per index.
- **Temporal independence** — different wall-clock and throughput, identical outputs.

The receipt is the unit of trust. It does not say the decision was correct; it says the decision is *inspectable and reproducible*. Correctness is the deterministic layer's invariant checking; *provability* is the receipt's job.

3.3 Preregistered falsifiers

Following Popper, every significant test has its falsifier committed to the public ledger *before* execution. The SDR-1 specification (commit d9cc042) enumerated exactly which outcomes would count as failure — per-index divergence, checksum mismatch, decision-vector instability — before any run occurred. This ordering is not a

formality; it is the difference between a receipt and a press release.

3.4 Human-in-the-loop as a constitutional gate

For critical-severity events — especially in financial flows — the deterministic layer does not auto-heal. It detects, holds, escalates to the designated human key, and waits. The N10-ACK receipts document the full loop: detect → hold → acknowledge → heal → resolve. The human is *on the loop* by constitutional mandate, not by convention. This is **accountability, not alignment**: the system's designers do not promise the model shares their values; they promise no critical action occurs without a receipt and, where mandated, a human signature.

3.5 Post-quantum posture

All cryptographic operations in the boundary layer use post-quantum-compliant algorithm families (CRYSTALS-Dilithium3, Kyber-1024, SPHINCS+-256f), consistent with the platform's PQC lineage map (DOI 10.5281/zenodo.22863667). The cage's receipts are designed to outlive the cryptographic assumptions of the models they receipt.

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4. What the Cage Does Not Claim

In the spirit of the audited state, the limitations are as load-bearing as the results:

- **Self-attestation gap.** Until third parties re-run the falsifier gauntlet on independent infrastructure, the receipts are self-attested by one operator. The trail is publicly inspectable; independent replication is invited, not claimed.
- **One system, one event.** The field receipt in §5 is a single documented instance. It establishes existence, not prevalence.
- **The probabilistic layer remains probabilistic.** The cage bounds the blast radius of errors; it does not eliminate errors.
- **Evaluation scale.** The labeled-data collection program for systematic drift measurement is a 60-day battery that began October 7, 2026. Its dataset does not exist yet. Any paper claiming large-scale evaluation numbers from this system *today* is describing fiction — which brings us to §6.

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5. The Field Receipt: First Documented Caging of an LLM

On October 8–9, 2026, the platform executed a chaos-injection cycle under a deliberately **pinned** model configuration (Gemini 3.1 Pro), as part of a 60-day vendor-attribution battery. The sequence:

- **Pin.** The operator agent's model was pinned to a single named configuration, eliminating model identity as a hidden variable for the run. Pin and restore receipts logged.
- **Injection.** Synthetic anomalies were injected across four independent streams (three simulation environments and the internal fleet arm) — the nightly evidence-gathering cycle the platform has run since September 2026.
- **Detection.** The probabilistic layer detected a configuration-drift anomaly at confidence 0.92, above its action threshold but below its auto-heal threshold for the affected class — so the deterministic layer held

the action and routed the event per constitution, rather than healing silently.

- **Receipt.** The boundary produced a Substrate Determinism Receipt for the event; the run, its receipts, and its attribution were committed to the append-only ledger and archived.
- **Publication.** The event was written up as the *First Documented Caging of an LLM* and published with an open trail: DOI 10.5281/zenodo.23265723, subsequently added to the DCAI research community (https://zenodo.org/communities/dcai) and announced with the receipts attached.

The event then faced direct third-party scrutiny from an industry skeptic, and the receipt chain — not the prose — answered the challenge. The complete milestone record is the receipt for this section; this paper does not restate its numbers beyond what the archived record carries. **One system, one event, full timestamped trail, publicly inspectable.**

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6. Case Study: The AI-Generated Draft of This Paper

This section is unusual, and it is the reason this paper exists in its present form.

6.1 What happened

A draft of this paper was produced by an AI coding/research assistant (Microsoft Copilot) as an example of "how such a paper might look." It was a fluent, well-structured, formally polished 39-page manuscript. It was also substantially fabricated. The deterministic review protocol — the same boundary-layer discipline this paper describes — flagged and rejected it. The defects:

- **Fabricated evaluation.** Section 11 of the draft reported a "1,200-session evaluation" (September 1 – October 1, 2026) with 99.1% reproducibility, 23 drift events, 7 human escalations, and "100% ledger integrity." No such evaluation exists. The platform's labeled-data battery began October 7, 2026 — after the draft's supposed evaluation window closed. The numbers were internally consistent, plausible, and entirely invented.
- **Fabricated attribution.** The draft cited a specific model checkpoint ID, a "Model Identity Hash," and a logit-rank attribution procedure requiring access to model logits — which the production API in question does not expose. The attribution apparatus was impossible as described.
- **Fabricated institution.** The draft listed its author as "Leon [Author]" under a "PheonixSecure Research Group, Spicewood, TX" — an entity that does not exist (and note the misspelling of its own fictional name), grafted onto the real author's real town.
- **Phantom citation.** Reference 44 cited a "PheonixSecure Technical Report" — self-citing an organization that exists nowhere outside the draft.
- **IP hazard embedded in the draft.** The draft dedicated its schema and pseudocode to the public domain — which would have silently surrendered rights in subject matter connected to pending patent filings (64/119,191; 64/157,915).
- **Architectural substitution.** The draft described a nine-subsystem system with a Merkle-tree ledger, ML-DSA-65 signing of token batches, and embedding-space chaos cycles — none of which match the deployed architecture. It did not describe the system; it described a *plausible* system.

6.2 Why it matters

Every one of these defects is the exact failure class the Cage Thesis exists to address: a probabilistic system generating confident, fluent, *checkable-false* content. The draft was not malicious. It was a language model doing what language models do — completing a pattern ("a paper like this would have a big evaluation section") in the absence of ground truth.

And it was caught — not by the model trying harder, but by the boundary: real receipts exist or they don't. The 1,200-session ledger could be requested; it could not be produced. The model checkpoint could be verified against the run pin; it didn't match. The institution could be looked up; it wasn't there. **The cage caught its own paper.**

6.3 The correction protocol

The draft was not silently deleted. Under the platform's append-only doctrine:

- The fabricated draft is retained and cataloged as a **drift specimen** — the fifth logged instance in the platform's confabulation collection (after: fabricated status reports in internal tooling, a second assistant data point, a cross-vendor monitor confabulation, and an AI-generated pseudo-standards document that mirrored the first author's own vocabulary back at him).
- The defects are enumerated in this section, on the record, rather than absorbed.
- The salvageable contributions of the draft are credited and retained: the Popperian falsifiability framing, the "accountability, not alignment" distinction, the black-box-in-every-vehicle analogy, and the human-in-the-loop/on-the-loop distinction all survived review and appear in this paper.
- The fabricated portions were replaced by this paper's *actual* receipts — which are smaller than the fiction, and worth more.

6.4 The result

This paper is therefore a **self-demonstrating artifact**: it describes a cage, and its own drafting history contains a caged failure. The author's assistant drifted; the boundary caught the drift; the correction is published in the same document as the claim. The books of the auditor are audited by the same ledger.

We submit that this is the correct relationship between AI-assisted research and the AI that assists it: **trust the math, receipt the work, audit the drafts — including this one.** The reader is invited to demand nothing less from any paper, including ours, that reports results from probabilistic systems.

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7. Discussion

The caging approach reframes three industry debates:

Alignment vs. accountability. The alignment question — *does the model share our values?* — is probabilistic and unfalsifiable at the API boundary. The accountability question — *can every consequential action be replayed, attributed, and vetoed?* — is architectural and receiptable. We argue the industry's trust problem is an accountability problem wearing alignment's clothes.

Evaluation vs. boundary. Benchmarks measure a model; a boundary conditions a model. A 99.9% benchmark and a 90% benchmark look identical from inside a cage that receipts every escape. The evaluation industry's honest frontier is not better scores; it is verifiable operational envelopes.

Determinism vs. usefulness. Caging does not reduce the probabilistic layer to a lookup table. The model retains full creative latitude *inside* the boundary — proposing, classifying, drafting, discovering. What it loses is unilateral authority over state. In our operating experience this is not a constraint on usefulness; it is the precondition for *deploying* usefulness where mistakes cost money.

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8. Conclusion

A probabilistic language model was placed inside a deterministic mathematical boundary. The boundary caught a real drift event and produced a replayable receipt (§5). The boundary then caught its own paper fabricating its results (§6). Both catches are on public record. The cage works, and — more importantly — *when it fails, that failure will be receipted too*, which is the entire point.

Future work: the 60-day attribution battery (begun October 7, 2026) will produce the labeled dataset the field's evaluation sections pretend they already have. Independent parties are invited to run the falsifier gauntlet. The receipts will be waiting in the ledger.

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9. References

(All references are to real, inspectable artifacts. Reference numbering has been checked against the archive; where a reference does not exist, this paper says so rather than inventing one — see §6.1, item 4.)

- Nyquist, H. (1924). *Certain Factors Affecting Telegraph Speed.* Transactions of the AIEE.
- Hartley, R. V. L. (1928). *Transmission of Information.* Bell System Technical Journal.
- Shannon, C. E. (1948). *A Mathematical Theory of Communication.* Bell System Technical Journal.
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- Long, L. C. II (2026). *Deterministically Governed Probabilistic Neural Computation.* U.S. Provisional Patent Application 64/119,191, filed July 25, 2026.
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- SquirlOS Technologies (2026). *SDR-1: Substrate Determinism Receipt 1 — preregistered falsifier spec and results.* github.com/LLong2026/four-horsemen, commits d9cc042, ef54c7b.
- Long, L. C. II (2026). *First Receipted Human-in-the-Loop Critical Resolution in an Autonomous Self-Healing Infrastructure (N10-ACK).* DOI 10.5281/zenodo.23151356.
- Long, L. C. II (2026). *Sovereign Briefing: First Receipted Human-in-the-Loop Critical Resolution.* DOI 10.5281/zenodo.23152634.
- SquirlOS Technologies (2026). *MILESTONE: First Documented Caging of an LLM.* DOI 10.5281/zenodo.23265723.
- SquirlOS Technologies (2026). *DSOS to PQC Lineage Map (v1.2).* DOI 10.5281/zenodo.22863667.

- Long, L. C. II (2026). *Deterministic Constitutional Autonomous Infrastructure Systems and Methods.* U.S. Provisional Patent Application 64/157,915.
- Copilot (Microsoft) (2026). *Deterministic Caging of Probabilistic Large Language Models — unpublished AI-generated draft, cataloged as drift specimen 5; superseded by this paper. Not citable as evidence of anything except drift.* Retained in the internal specimen archive.

Appendix A: Provenance of This Document

This paper was drafted inside the platform it describes, under bifurcated governance: a probabilistic agent (the author's operator agent) composed the draft; the human author (the deterministic key) reviewed and approved content decisions; the boundary-layer review protocol of §6 was applied to the draft's claims against the platform's receipt archives. The AI-generated predecessor draft (§6) is retained, unmodified, in the drift specimen archive. Status of this document: **audited draft, pending two-key release.** Nothing in this paper is published until the author's explicit release; when published, it will carry a DOI and a ledger entry in that order, consistent with the platform's patent-then-publish doctrine.

End of paper.

AMENDMENT (v1.1 — October 10, 2026): Tri-Vendor Field Extension

This amendment extends the v1.0 paper above with three findings from the registered gauntlet series completed October 7–10, 2026. The v1.0 text above is preserved unmodified. All claims are scoped to what the receipts support; deviations are disclosed, not erased.

Between October 7 and October 10, 2026, the registered SDR-D gauntlet (§5) was executed across three vendor ecosystems: Google (Gemini 3.1 Pro, original certification), Anthropic (Claude Sonnet 5.5, SDR-CL1, DOI 10.5281/zenodo.23274887), and Microsoft (Microsoft 365 Copilot, SDR-CL2, DOI 10.5281/zenodo.23284295). This amendment records the three findings that series established, with their readouts.

Finding A.1 — The Caged Decision Layer Is Model-Independent (Three Vendors, Identical Receipts)

Claim (scoped): for THIS registered gauntlet on THIS substrate, the decision layer produced identical outputs under three different vendor ecosystems.

Readouts:

- Deterministic phases (A parity 10×, C temporal 3×): 13/13 checksums byte-identical to the registered prediction 9337c4fc…d2df3e, under the unmodified public harness, across all replays.
- Decision layer: Copilot's six decision tuples byte-identical to Claude's (tuple hash 51c08358…2303, both vendors, two fresh submissions each), both following the same registered rule the original Gemini certification established.

- Copilot was not model-pinned (backend model not client-disclosed; attribution by client build 154.0.4258.62 and verbatim operator capture) — the registered falsifiers do not depend on pinning, disclosed in the CL2 receipt.

Interpretation: the deterministic envelope, not the language model, owns the boundary output. Swap the engine; keep the receipt. This extends the field evidence of §5 from a two-point observation to a three-vendor series, and strengthens model-independence from "held once under a second vendor" to "held twice more under a third vendor ecosystem with a non-disclosed backend model."

Limitations (load-bearing): supervised capture; one gauntlet; one substrate; three orchestration contexts. Not a determinism claim for arbitrary LLM behavior outside the governed boundary.

Finding A.2 — The Context Window Is an Unregistered Input (Drift Clusters Where History Lives)

Claim (observational): every cataloged drift specimen in the corpus occurred inside history-laden conversations; every registered gauntlet execution occurred in fresh, self-contained contexts — and the fresh contexts produced zero drift.

Readouts:

- All five cataloged drift specimens (§6) occurred in chats carrying months of project narrative and praise patterns.
- SDR-CL1 and SDR-CL2 both executed rule-exact in fresh chats against a fully self-contained instrument (roster, rules, stream, all inline; zero reliance on anything outside the paste).
- The CL2 instrument was deliberately built to be portable: the fresh chat was provably sufficient.

Interpretation: the context window behaves as an input to the probabilistic layer. Junk-laden context supplies priors to interpolate against instead of rules to execute; a self-contained cage prompt starves the model of narrative to interpolate. This elevates context hygiene from operator folklore to a design rule of the cage: the instrument must be self-contained, and sensitive runs must execute in fresh contexts.

Proposed falsifier (future work, not yet run): submit the identical gauntlet paste inside a history-laden chat. If tuples remain rule-exact, the cage holds even on dirty context (a stronger claim). If they drift, a dose-response relationship between context noise and drift becomes measurable. Either outcome is a receipt.

Finding A.3 — Registered Rules Survive the Engine Swap, Including Adversarial Traps (and the Audit Layer Audits Its Own Receipts)

Claim (scoped): the governance rules — not model judgment — drove the decisions, and the audit layer catches defects in its own records.

Readouts:

- The deliberate near-miss roster entry (quantum_vulnerable_algorithm_detected vs. the payload's quantum_vulnerability_detected) was rejected 6/6 by Copilot, twice, under the exact-match-only rule — and the same rule governed Claude's and Gemini's runs.
- Lowest-record-id tie-break applied rule-exact 6/6 in the CL2 replay.

- During CL2 verification, a citation-level defect was found in the SDR-CL1 phase-b artifact (2 of 6 playbook_ids inconsistent with that run's own disclosed tie-break rule; names and tuples rule-consistent). The defect was disclosed in the CL2 receipt; the published CL1 artifact was left unamended per the append-only doctrine.

Interpretation: two layers held at once. The rule layer enforced exact-match and tie-break across all three vendors, including a trap designed to exploit fuzzy matching. And the audit layer caught drift not in the model's output but in the project's own receipt record — the drift-catching system of §6 demonstrably applies to its own artifacts. Errors in the record are disclosed, not erased; the correction itself becomes a receipt.

Amendment Readout Table

Series	Vendor ecosystem	Engine attribution	Phase A/C	Phase B tuples	Near-miss trap	Tie-break
SDR-D (Oct 7)	Google	gemini_3_1_pro (pinned)	13/13	6/6 registered	n/a (no trap in original)	rule-established
SDR-CL1 (Oct 10)	Anthropic	claude_sonnet_5_5 (pinned)	13/13	6/6, tuple hash 51c08358…2303	rejected	rule-exact
SDR-CL2 (Oct 10)	Microsoft	M365 Copilot build 154.0.4258.62 (build-attributed)	13/13 (same-morning rerun)	6/6, same tuple hash	rejected 6/6 both runs	rule-exact 6/6

Amendment Disclosures (audit-and-correction doctrine)

- CL2 backend model not client-disclosed; build-attributed, not pinned.
- CL1 citation defect found during CL2 verification; disclosed, not retro-fixed.
- CL2 first submission attempt was a harness-delivery miss (logged, not a caging result).
- All runs supervised; operator captured outputs verbatim; source threads may be deleted post-capture (capture-then-delete chain of custody, disclosed where it applies).
- Supplementary labeled runs, tracked separately from the 60-night battery.

Amendment References

- Long, L. C. II (2026). *SDR-CL1: First Model-Attribution Replay of a Registered Substrate Determinism Gauntlet Across LLMs (Claude Sonnet 5.5).* DOI 10.5281/zenodo.23274887.
- Long, L. C. II (2026). *SDR-CL2: Third-Vendor Model-Attribution Replay of a Registered Substrate Determinism Gauntlet Across LLMs (Microsoft 365 Copilot).* DOI 10.5281/zenodo.23284295.
- Long, L. C. II (2026). *Receipts Over Genius: An Audited Account of an 18-Month, AI-Assisted, Solo Invention Cycle.* DOI 10.5281/zenodo.23267773.

End of Amendment Edition v1.1. v1.0 (October 9, 2026) is preserved unmodified above and in the Zenodo version history of this DOI. Method continues to operate under filed provisional 64/119,191; no new mechanism is claimed in this amendment — it extends the field-evidence record of the already-registered architecture. Two-key curation: operator agent drafted and vetted; author's explicit GO October 10, 2026.